Bangladesh National Building Code (BNBC) 2020

PART - I
SCOPE AND DEFINITIONS

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রেজিস্টার্ড নং ডি এ-১ "জাতির পিতা বঙ্গবন্ধু শেখ মুজিবুর রহমানের জন্মশতবার্ষিকী উদ্‌যাপন সফল হোক"

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কর্তৃপক্ষ কর্তৃক প্রকাশিত

বৃহস্পতিবার ফেব্রুয়ারি ১১, ২০২১

Government of the People's Republic of Bangladesh
Ministry of Housing and Public Works

Notification

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S.R.O. No.55-Law/2020.—In exercise of the powers conferred under section 18A of the Building Construction Act, 1952 (Act No. II of 1953) the Government is pleased to make the following Code by repealing the Bangladesh National Building Code, 2006, namely :—

PART I - SCOPE AND DEFINITIONS
Chapter 1
Title, Purpose, Scope, Etc

1. **Title and Commencement**.—(1) This Code may be called the Bangladesh National Building Code (BNBC) 2020.

(2) It shall come into force at once.

2. **Purpose.—**(1) The purpose of this Code is to establish minimum standards for design, construction, quality of materials, use and occupancy, location and maintenance of all buildings within Bangladesh in order to safeguard, within achievable limits, life, limb, health, property and public welfare.

(2) The installation and use of certain equipment, services and appurtenances related, connected or attached to such buildings are also regulated herein to achieve the same purpose.

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(3) The expressed intent of this Code is to ensure public safety, health and general welfare insofar as they are affected by the construction, alteration, repair, removal, demolition, use or occupancy of buildings, structures or premises, through structural strength, stability, means of egress, safety from fire and other hazards, sanitation, light and ventilation.

3. Scope.—(1) The provisions of this Code shall apply to the design, construction, use or occupancy, alteration, moving, demolition and repair of any building or structure and to any appurtenances installed therein or connected or attached thereto, except such matters as are otherwise provided for in other laws controlling and regulating buildings.

(2) If for any case different sections of this Code provide different specifications for materials, methods of design or construction, or other requirements, the most restrictive specification shall govern.

(3) In case of any conflict between a general requirement and a specific requirement, the specific requirement shall prevail.

(4) Unless otherwise explicitly stated in this Code, all references to part, chapter or section numbers or to provisions not specifically identified by number, shall be construed to refer to such part, chapter, section or provision of this Code.

(5) References made to a section without mentioning a part shall be construed to refer to that section of the part in which the reference is made.

(6) The provisions of any appendix in this Code shall not be mandatory unless they are referred to as such in any section of the Code or they are specifically adopted by any regulation.

(7) Inspection conducted or permission granted for any building or plan of building, under the provisions of this Code, shall not be construed as a warranty of the physical condition of such building or the adequacy of such plan.

(8) Neither the Authority nor any employee thereof shall be liable for damages or any defect or hazardous or illegal condition or inadequacy in such building or plan, nor for any failure of any component of such building which may occur subsequent to such inspection or granting of permission under the provisions of the Code.

4. **Existing buildings**.—(1) Buildings which are in existence on the date of commencement of this Code may have their use or occupancy continued without undergoing any alteration, abandonment or removal unless in the opinion of the Authority such continued use is hazardous to life and property and provided such use or occupancy was legal on the date of commencement of this Code.

(2) Buildings approved before commencement of this Code and compliant under the repealed Code may continue to be used or occupied unless any deviation is made thereafter or any deterioration has rendered the building unsafe in the opinion of the Authority.

(3) Additions, alterations, modifications or repair to an existing building may be made without requiring the existing building to comply with all the requirements of this Code, provided the additions, alterations, modifications or repairs conform to that required for a new building and such additions or alterations shall not be permitted when the existing building is not in full compliance with the provisions of this Code except when the addition or alteration will result in the existing building or structure being no more hazardous based on life safety, fire safety and sanitation than it was before the addition or alteration was undertaken.

(4) Any building together with the new additions shall not exceed the height, number of storeys and area specified in this Code for new buildings having the relevant occupancy and type of construction.

(5) Non-structural alterations or repairs to an existing building or structure which do not adversely affect any structural member, nor reduce the strength of any part of the building or structure to result in an unsafe condition shall be made with materials and components having the required fire resistance.

(6) Change in use or occupancy in an existing building may be made when such change complies with the requirements of this Code for a new building and provided such change does not render any part or the whole of the affected building or structure any more hazardous based on life safety, fire safety and sanitation than it was before such change was effected.

5. Historic or architecturally valuable buildings.—A building or structure which has been designated by official action as having special historical or archaeological interest, or a building or structure identified by a legally constituted authority as being architecturally valuable, may be undertaken for repairs, alterations and additions necessary for its preservation, restoration, rehabilitation or continued use, provided :

(a) the proposed repair, alteration or addition to buildings of historical or archaeological significance is approved by the legally constituted authority, such as the Department of Archaeology;

(b) the proposed repair, alteration or addition to buildings of architectural value does not impair the aesthetic quality and architectural character of such buildings; and

(c) the restored building or structure will be no more hazardous, if any, based on life safety, fire safety and sanitation than the existing building.

PART I
Chapter 2
Definitions

6. Definitions.—In this Code, unless there is anything repugnant in the subject or context,—

ACCESSORY USE	means any use subordinate to the major use which is normally incidental to the major use.
ALTERATION	means any change, addition or modification in construction such as structural, dimensional, or any removal of any part of a building or any change to or closing of any required means of ingress or egress or a change to the fixtures or equipment or any change in land use or occupancy or use.
APPLICANT	means a person, a firm, a company, a corporation, or a government, semi-government or non-government agency who intends to undertake any work regulated by this Code and who has filed an application to the Building Official for this purpose in a form prescribed in the Code.
APPROVED	means approved by the Authority.
APPROVED PLAN	means the set of plans, designs and specifications of building submitted to the Authority as per provision of this Code and duly approved and sanctioned by the Authority.
ARCHITECT	means a person who has a Bachelor Degree in Architecture and is a member of the Institute of Architects, Bangladesh (IAB).
AUTHORITY	means the Bangladesh Building Regulatory Authority.
AUTHORIZED OFFICER	means BUILDING OFFICIAL.
BASEMENT	means a floor of a building more than 50 percent of which is situated at a depth of 1 m or more below crown of the main entry road.
BUILDING	means any permanent or semi-permanent structure which is constructed or erected for human habitation or for any other purpose and includes but not limited to the foundation, plinth, walls, floors, roofs, stairs, chimneys, fixed platform, verandah, balcony, cornice, projections, extensions, annexes etc. The term building will also include the sanitary, plumbing, electrical, HVAC, appurtenances and all other building service installations which are constructed or erected as an integral part of a building.

BUILDING LINE	means the line up to which the plinth of a building may lawfully extend. Also known as SETBACK LINE.
BUILDING OFFICIAL	means a person who is the jurisdictional administrator of this Code appointed by the Authority.
COMMITTEE	means a Building Construction Committee constituted for any area in the prescribed manner, if necessary.
CONSTRUCT	means ERECT.
CONVERSION	means the change in occupancy or premises to any occupancy or use requiring new occupancy permit.
COVERED AREA	means the ground area above the plinth level which is covered by a building structure. The covered area of a building shall exclude gardens, wells, cornice, sunshade, pergola, septic tank, soak well, unpaved uncovered water body, fountains, drainage structures, boundary wall, gates, porch, uncovered staircase, watchman's cabin, detached pump house, garbage chutes and other uncovered utility structures.
DEVELOPMENT	means carrying out construction of buildings, engineering, mining or other operations in, or over or under land or water. Includes re-development and layout and subdivision of any land. 'To develop' and other grammatical variations shall be interpreted accordingly.
DIPLOMA ARCHITECT	means a person who has a Diploma in Architecture from any recognized Polytechnic or Technical Institute and is a member of the Institution of Diploma Engineers, Bangladesh (IDEB).
DIPLOMA ENGINEER	means a person who has a Diploma in Engineering from any recognized Polytechnic or Technical Institute and is a member of the Institution of Diploma Engineers, Bangladesh (IDEB).
DRAIN	means a conduit or channel for conveying water, sewage, or other waste liquid for subsequent disposal.
DRAINAGE	means the disposal of any liquid with a system meant for this purpose.
ENGINEER	means a person who has a Bachelor Degree in Engineering and is a member of the Institution of Engineers, Bangladesh (IEB).

ERECT	means to erect a new building or re-erect an existing building or to convert a building from one occupancy to another. Also known as CONSTRUCT.
FORMATION LEVEL	means finished ground level of a plot. For hilly areas formation levels shall be the gradient of the plot surface.
GEOTECHNICAL ENGINEER	means engineer with Masters degree in geotechnical engineering having at least 2 (two) years of experience in geotechnical design/construction or graduate in civil engineering/engineering geology having 10 (ten) years of experience in geotechnical design/construction.
ENGINEERING GEOLOGIST	means a person having a postgraduate degree in engineering geology and having 2 years of experience in geotechnical exploration and interpretation.
GOVERNMENT	means the government of the People's Republic of Bangladesh.
GRADE	means the lowest point of elevation of the finished surface of the ground, pavement or footpath within the area between the building and the property line or a line 1.5 m from the building whichever is nearer the building.
HEIGHT OF BUILDING	means the vertical distance from a reference datum to the highest point of the building which includes all building appurtenances like overhead water tank, machine room, communication tower etc. The reference datum shall be the elevation of the nearest footpath or the elevation of the nearest road or street or public way at its centre line, whichever is higher.
HIGH RISE BUILDING	means any building which is more than 10-storey or 33 m high from reference datum. Building appurtenances like overhead water tank, machine room, communication tower etc. will not be considered in determining the height.
OCCUPANCY or USE GROUP	means the purpose for which a building or a part thereof is used or intended to be used.
OCCUPANCY, MAJOR	means the major or principal occupancy of a building or a part thereof which has attached to it subsidiary occupancy or occupancies contingent upon it.

OCCUPIER	means a person paying or liable to pay rent or any portion of rent of a building in respect of which the word is used, or compensation or premium on account of occupation of such building and also a rent-free tenant. Does not include a lodger and the words 'occupancy' and 'occupation' do not refer to the lodger. In such cases, the owner himself or herself is living in his or her own building, he or she shall be deemed to be the occupier thereof.
OWNER OF A BUILDING	means the person, organization or agency at whose expenses the building is constructed or who has the right to transfer the same and includes his or her heirs, assignees and legal representatives, and a mortgagee in possession.
PERMIT	means a written document or certificate issued by the Authority for carrying out a specific activity under the provisions of this Code.
PLANNER	means a person who has a Bachelor or a Postgraduate Degree in Planning and is a member of the Bangladesh Institute of Planners (BIP).
PLINTH AREA	means the elements from the building bases which are exposed above the formation level to form a covered floor area by joining the peripheral points of the elements which are intersected at finished floor plane at the height of plinth level.
PLINTH LEVEL	means height of a covered finished floor which is not more than 1 m above the formation level nor 1.85 m from the crown of adjacent road level.
PLOT	means SITE.
PLUMBING ENGINEER	means an Engineer (Civil/Mechanical) who has experience in the field of plumbing or sanitation.
PUBLIC WAY	means ROAD.
RELIABLE LITERATURE	means RELIABLE REFERENCE.
RELIABLE REFERENCE	means reference materials such as published article, codes, standards or other material judged to be reliable by the professional users and specialists in the subject concerned. This may also be referred to as RELIABLE LITERATURE.

ROAD	means a thoroughfare or public way which has been dedicated or deeded to the public for public use and also known as STREET.
ROAD LINE	means a line defining the side limits of a road.
ROOM HEIGHT	means the clear head room between the finished floor surface and the finished ceiling surface or the underside of the joists or beams, whichever is lower.
SANCTIONED PLAN	means the set of plans, design and specifications of a building submitted to the Authority as per provision of this Code and duly approved and sanctioned by the Authority.
SERVICE ROAD	means a road or lane provided at the rear or side of a plot for service purposes.
SETBACK LINE	means BUILDING LINE.
SITE	means a piece or parcel of land on which a building is intended to be or has already been constructed and also known as PLOT.
SPECIALIST	means a professional who by education, research, practice and experience specializes in a particular branch of a broader discipline and is generally judged to be so by the professional body in the relevant discipline.
STOREY	means the portion of a structure between tops of two successive finished floor surfaces and for the topmost storey, from surface of the finished floor of topmost floor to the top of the roof above.
STOREY, FIRST	means the lowest storey in a building which qualifies as a storey as defined herein; for a building with a basement, it is the storey just above the basements.
STREET	means ROAD.
STREET LEVEL	means the elevation of the centre line of any road or street which a plot fronts.
STREET LINE	means ROAD LINE.
SUPERVISOR, CONSTRUCTION	means an Architect or Engineer or Diploma Architect or Diploma Engineer having experience in supervision of construction works.
UNSAFE BUILDING	means a building which, in the opinion of the Building Official, is structurally unsafe, or insanitary, or lacks proper means of ingress or egress, or which constitutes a hazard to life or property.

PART I
Chapter 3
Abbreviations

7. Abbreviations of names and words.—(1) Names of institutions, organizations and professional societies referred to in this Code are listed below in an alphabetical order, namely :—

ACI	American Concrete Institute; Box 19150, Redford Station, Detroit, MI 48219, USA.
AISC	American Institute of Steel Construction, Inc.; 400 North Michigan Avenue, Chicago, IL 60611, USA.
AISE	Association of Iron and Steel Engineers; Suite 2350, Three Gateway Center, Pittsburgh, PA 15222, USA.
AISI	American Iron and Steel Institute; Suite 300, 1133 15th Street N.W., Washington, DC 20005, USA.
ANSI	American National Standards Institute; 1430 Broadway, New York, NY 10018, USA.
ASHRAE	American Society of Heating, Refrigerating and Air-conditioning Engineers, Inc.; 345 East 47th Street, New York, NY 10017, USA.
ASME	American Society of Mechanical Engineers; United Engineering Centre, 345 East 47th Street, New York, NY 10017, USA.
ASTM	American Society for Testing and Materials; 1916 Race Street, Philadelphia, PA 19103, USA.
AWS	American Welding Society; 550 N.W. LeJeune Rd., P.O. Box 351040, Miami, FL 33135, USA.
BIP	Bangladesh Institute of Planners, Planner's Tower (Level-7), 13/A, Bir Uttam C.R. Datta (Sonargaon) Road, Bangla Motor, Dhaka-1000, Bangladesh.
BOCA	Building Officials and Code Administrators International Inc.; 1313 East 60th Street, Chicago, IL 60637, USA.
BPDB	Bangladesh Power Development Board; WAPDA Building, Motijheel Commercial Area, Dhaka-1000, Bangladesh.
BSI	British Standards Institution; 2 Park Street, London W1A 2BS, UK.
BSTI	Bangladesh Standards and Testing Institution; 116A Tejgaon Industrial Area, Dhaka-1208, Bangladesh.
BWDB	Bangladesh Water Development Board; WAPDA Building, Motijheel Commercial Area, Dhaka-1000, Bangladesh.

CDA	Chittagong Development Authority; Station Road, Chittagong, Bangladesh.
CGSM	Canadian General Standards Board; Technical Information Unit, Ottawa, CANADA K1A 1G6.
DOA	Department of Architecture; Sthapatya Bhaban, Shahid Capt. Mansur Ali Sarani, Segunbagicha, Dhaka-1000, Bangladesh.
DPHE	Department of Public Health Engineering; DPHE Bhaban, 14, Shaheed Captain Mansur Ali Sarani, Kakrail, Dhaka-1000, Bangladesh.
EED	Education Engineering Department; Shikkha Bhaban, Dhaka-1000, Bangladesh.
HED	Health Engineering Department; Ministry of Health and Family Welfare, 105-106, Motijheel C/A, Dhaka-1000, Bangladesh.
FM	Factory Manual; Standards Laboratories Department, 1151 Boston Providence Turnpike, Norwood, MA 02062, USA.
FSCD	Fire Service and Civil Defence, Kazi Alauddin Road, Dhaka-1000, Bangladesh.
HBRI	Housing and Building Research Institute, 120/3, Darus-Salam, Mirpur, Dhaka, Bangladesh.
IAB	Institute of Architects Bangladesh, Plot-11, Block-E, Road-7, Sher-e-Bangla Nagar, Agargaon, Dhaka.
IEB	The Institution of Engineers, Bangladesh, Ramna, Dhaka-1000.
IDEB	Institution of Diploma Engineers, Bangladesh, IDEB Bhaban, 160/A, Kakrail VIP Road, Dhaka-1000.
ICBO	International Conference of Building Officials, 5360 South Workman Mill Road, Whittier, CA 90601, USA.
ISO	International Organization for Standardization, 1, Rue de Varembé, Case Postal 56, CH-1211, Genève 20, Switzerland.
ISSMFE	International Society of Soil Mechanics and Foundation Engineering, University Engineering Department, Trumpington St, Cambridge CB21PZ, UK.
KDA	Khulna Development Authority, Shib Bari Crossing, Khulna-9100, Bangladesh.
LGED	Local Government Engineering Department, LGED Bhaban, Sher-e-Bangla Nagar, Agargaon, Dhaka-1207. Bangladesh.

NFPA	National Fire Protection Association, Batterymarch Park, Quincy, MA 02269, USA.
NHA	National Housing Authority, Grihayan Bhaban, 82, Segunbagicha, Dhaka, Bangladesh.
PWD	Public Works Department, Purto Bhaban, Shahid Capt. Mansur Ali Sarani, Segunbagicha; Dhaka-1000, Bangladesh.
RAJUK	Rajdhani Unnayan Kartripakkha, Rajuk Avenue, Motijheel, Dhaka-1000, Bangladesh.
RCSC	Research Council on Structural Connections of the Engineering Foundation, American Institute of Steel Construction (AISC).
RDA	Rajshahi Development Authority, Rajshahi-6203, Bangladesh.
RMA	Rubber Manufacturing Association, 1400 K Street N.W., Washington, DC 20005, USA.
SBCCI	Southern Building Code Congress International, 3617 8th Ave, S. Birmingham, AL 35222, USA.
SMACNA	Sheet Metal and Air Conditioning Contractors' National Association, 8224 Old Courthouse Road, Tysons Corner, Vienna, VA 22180, USA.
SPRI	Single Ply Roofing Institute, 104 Wilmont Road, Suite 201, Deerfield, IL 600015-5195, USA.
UDD	Urban Development Directorate, Ministry of Housing and Public Works, 82, Segunbagicha, Dhaka-1000, Bangladesh.
UL	Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611, USA.

(2) The abbreviations of words used in this Code are listed below in an alphabetical order. Abbreviations not explicitly defined herein below shall be construed to have their usual meaning as the context implies.

BDS	Bangladesh Standards; published by the BSTI
BNBC	Bangladesh National Building Code; published by HBRI
BS	British Standard; published by the BSI
CBF	Concentric Braced Frame
CFC	Chlorofluorocarbon

CGI	Corrugated Galvanized Iron
CWPC	Cold Drawn Low Carbon Wire Prestressed Concrete
DCP	Dry Chemical Powder (fire extinguisher)
DDT	Dichlorodiphenyltrichloroethane
DPC	Damp-proof Course
EBF	Eccentric Braced Frame
FAR	Floor Area Ratio
FM	Fineness Modulus
FPA	Flood Prone Area
GI	Galvanized Iron
IBC	International Building Code
IMRF	Intermediate Moment Resisting Frame
IS	Indian Standard; published by the Bureau of Indian Standards
LFD	Load Factor Design
LPG	Liquefied Petroleum Gas
MCSP	Multipurpose Cyclone Shelter Program
OMRF	Ordinary Moment Resisting Frame
RC	Reinforced Concrete
RS	Rolled Steel
RSJ	Rolled Steel Joist
SMRF	Special Moment Resisting Frame
SPA	Surge Prone Area
SRSS	Square Root of the Sum of the Squares
UBC	Uniform Building Code; published by the ICBO
WSD	Working Stress Design
cps	Cycles per second